

- Coordinating all the tasks to meet learning objectives

Building learning resources

Learning resources form the core of any teaching-learning process. Learning content and how it is delivered will determine the success or failure of the teaching-learning process. Though there are various types of learning resources, for the sake of simplicity, let us classify the resources that a teacher can build for remote learning into the following categories:

- Presentations
- Illustrations and mind maps
- Video lectures
- Podcasts
- Interactive content

Let's explore the tools belonging to each of these categories.

Presentations: Presentations are the most used method of delivering content. The popular open source tools to build presentations are listed below:

- Impress (Libreoffice)
- Beamer (LaTeX)
- Reveal.js

If you are already using some sort of proprietary presentation software then you might find shifting to LibreOffice Impress very simple and effective.

Beamer is LaTeX's tool for building presentations. If you are teaching a subject that involves

equations, mathematical notations or algorithms, then Beamer will make your presentations look elegant and professional.

If you want to build browser based presentations, then you should give Reveal.js a try. If you have introductory knowledge of HTML, you can build presentations that run inside the browser using Reveal.js. A section of the code customised from the official demo and its output (Figure 3) is shown below:

```
<div class="slides">
<section>
<h2>Open Source For You </h2>
<h3>Presentation on Remote Learning Tools </h3>
<p>
<small>Created by <a href="http://kskuppasamy.in">Dr. K.S. Kuppasamy </a> </small>
<small>Created using <a href="">reveal.js </a> </small>
</p>
</section>
<section>
<h2>Reveal.js</h2>
<p>
Reveal.js makes the presentation building so simple and effective.
</p>
</section>
```



Figure 2: Various categories of resources



Figure 3: Reveal.js browser based presentation

There are powerful Web based presentation frameworks such as Impress.js. If you want to check out Web based presentations, try exploring <https://github.com/impress/impress.js>.

Mind maps: Mind maps make explanations more effective. There are many mind-mapping tools available like FreeMind, FreePlane and Xmind (Figure 4). Xmind provides many features and makes the process simple.

Creating a video lecture: Video lectures are an important component of remote teaching. The sequence of steps involved in building a screen-casting based video lecture is as follows:

1. Prepare the presentation slides (using Impress, Beamer or Reveal.js).
2. Use a screen-recording tool to record the slides along with the voice-over (Open Broadcaster Software – OBS).
3. Use audio-editing tools to enhance your audio recording (Audacity).
4. Use video-editing tools to edit your video (OpenShot, Kdenlive, Shotcut).
5. Upload the lecture to the Web or LMS to share it with your students.

Step 1: This process is already explained in the 'Presentation' section of this article.

Step 2: This step involves using screen recording software like Open Broadcaster Software (OBS), which is a powerful tool. Indeed, it's not only for screen recording but also has powerful streaming capabilities that enable users to set up various scenes, sources, etc. You can include the inputs through a webcam, screen contents, microphone, etc. In my opinion, everyone who wants to create a video lecture must spend time on exploring OBS.

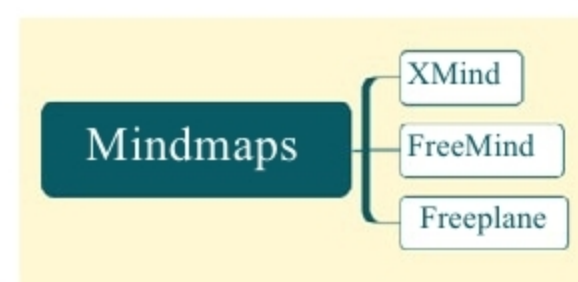


Figure 4: Mind mapping